

LESSON 5-2 • SESSION 1 • PRACTICE • FORM A**Area of Triangles**

Learning target – I can find the area of a triangle using one half the base times the height. · 6.G.A.1

REMEMBER

- Two matching triangles make a parallelogram, so a triangle is **half**: $A = \frac{1}{2} \times b \times h$.
- The **height** makes a square corner with the base.
- In a **right triangle**, the two legs at the square corner are the base and height.

WORKED EXAMPLE

A triangle has a base of 14 cm and a height of 9 cm. Find its area.

Step 1 $14 \times 9 = 126$ **Step 2** 126 is the parallelogram made by two triangles.**Step 3** $126 \div 2 = 63$ **Answer** 63 cm^2 **WORDS** right triangle (*triángulo rectángulo*) · height (*altura*) · congruent (*congruente*)**PRACTICE**

show your work · label units

1. Find the area of each triangle with base b and height h .

a. $b = 12 \text{ cm}$, $h = 5 \text{ cm}$ _____

b. $b = 14 \text{ in}$, $h = 3 \text{ in}$ _____

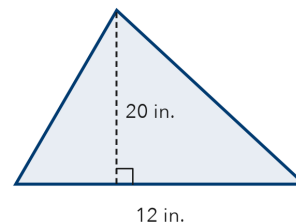
c. $b = 20 \text{ ft}$, $h = 7 \text{ ft}$ _____

d. $b = 11 \text{ m}$, $h = 8 \text{ m}$ _____

2. **Pennant Flag.** Two copies of this flag fit together to make a parallelogram.

a. Area of the parallelogram _____ in^2

b. Area of one flag _____ in^2



3. **Cut Sandwich.** A sandwich is cut on a diagonal into two equal triangles. Each has a base of 10 cm and a height of 8 cm.

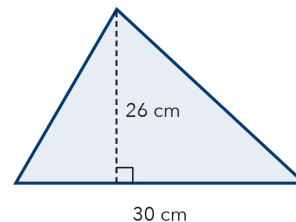
a. Area of one triangle _____ cm^2

b. Whole sandwich _____ cm^2

4. **Right-Triangle Sail.** A sail is a right triangle. Its two legs are 6 ft and 11 ft. Find its area.

Hint The legs meet at a square corner.**Answer** _____ ft^2

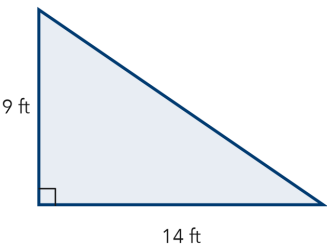
5. **Warning Sign.** Find the area of the sign.

Hint Multiply, then take half.**Answer** _____ cm^2 

6. **Recess Wedge.** A wedge of the recess yard is a right triangle. Find its area.

Hint In a right triangle, the legs are the base and height.

Answer _____ ft²



7. **Two Banners.** Banner P is a triangle with base 18 m and height 10 m. Banner Q has base 12 m and height 13 m. Find each area. Which covers more space?

Show your work.

Areas, which _____

8. **Garden Plot.** A triangular garden has a base of 24 ft and a height of 15 ft. One bag of mulch covers 30 ft² and costs \$6.

- a. Find the area. _____ ft²
 b. How many bags are needed? _____
 c. Find the total cost. What happens to the cost if the base is doubled? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Andre's Work.** A triangle has a base of 22 cm and a height of 10 cm. Andre wrote: " $A = 22 \times 10 = 220 \text{ cm}^2$." Explain his mistake. Then find the area.

TEST PRACTICE

choose the best answer

10. A triangle with base 14 in. and height 6 in. has area –
A 20 in² **B** 42 in² **C** 84 in² **D** 168 in²
11. The area of a triangle is what fraction of a rectangle with the same base and height?
A $\frac{1}{4}$ **B** $\frac{1}{3}$ **C** $\frac{1}{2}$ **D** 2

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

120 H	33 L	126 Z	780 Q	66 X	63 I	180 N	390 V	40 A	\$36 G
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#2b	#3a	#4	#5	#6	#8a	#8c
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____